

 Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)		COURSE SYLLABUS	
SCHOOL OF COMPUTER ENGINEERING		W.E.F	AY: 2024- 2025 (Rev. 2022)
FINAL YEAR BACHELOR OF TECHNOLOGY COMPUTER ENGINEERING		COURSE NAME	Generative Artificial Intelligence
		COURSE CODE	2304422T
		COURSE CREDITS	3
RELEASE DATE : 01/07/2025		REVISION NO.	0.0

TEACHING SCHEME (HOURS/WEEK)		EXAMINATION SCHEME AND MARKS					
		THEORY			LABORATORY		TOTAL
LECTURE	PRACTICAL	IA	MSE	ESE	CA	PRACT/DEMO/PRES.	
3	NIL	40	NIL	60	NIL	NIL	100

PREREQUISITE KNOWLEDGE : Machine Learning, Deep Learning.

COURSE OBJECTIVES :

- 2304422T.CEO.1: Understand principles of generative models.
- 2304422T.CEO.2: Develop practical skills in building and deploying generative AI models.
- 2304422T.CEO.3: Analyze ethical considerations and impacts of generative AI in different fields.
- 2304422T.CEO.4: Explore advanced models such as GANs, VAEs, Diffusion Models, and Transformer-based architectures.

COURSE OUTCOMES :

- After successful completion of the course, students will be able to,
- 2304422T.CO.1: Develop Generative AI models and diffusion models to generate data across various domains like images, text, and music [L3].
- 2304422T.CO.2: Use industry-standard tools and platforms for building, deploying, and scaling generative AI models [L3].
- 2304422T.CO.3: Analyze the performance of Generative AI models for real world applications [L4].
- 2304422T.CO.4: Determine the ethical challenges posed by generative AI technologies, including bias, misinformation, and intellectual property concerns [L5].

COURSE ABSTRACT:

This advanced course in Generative AI explores the theoretical underpinnings and practical implementations of AI models capable of generating original content. Leveraging foundational knowledge from Artificial Intelligence and Machine Learning and Deep Learning, students will delve into the architecture, training, and application of generative models, including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Transformer-based models such as GPT.

Through hands-on projects and case studies, students will learn to design, implement, and evaluate generative models for tasks such as image synthesis, text generation, and style transfer. Emphasis is placed on understanding the ethical implications and potential biases of generative AI, as well as methods to mitigate risks in applications. By the end of the course, students will have practical expertise in deploying generative models and insights into emerging applications across industries.

THEORY COURSE CONTENTS**UNIT 1 Introduction to Generative AI****08 HOURS**

Applications/Case Study: AI-Generated Music:Open AI's Jukebox, Text Generation:AI Dungeon
Contents: Overview of AI and generative models. Difference between discriminative and generative models. Applications of generative models (art, music, image generation, etc.). Historical context of generative models. Large Language Models.

Further Readings: Interactive Generative Systems

UNIT 2 Autoencoders and Variational Autoencoders (VAE)**09 HOURS**

Applications/Case Study: Medical Imaging, Anomaly Detection, Customer Segmentation and Recommendations

Contents: Introduction to Autoencoders (AE): Encoder, Decoder, Applications (Data Compression, Denoising). Variational Autoencoders (VAE): KL divergence, reparameterization trick, latent space exploration.

Self Study: Limitations of AEs and VAEs (overfitting, interpretability)

UNIT 3 Generative Adversarial Networks (GANs)**08 HOURS**

Applications/Case Study: Synthetic Medical Imaging Data, Deepfake Creation and Detection.

Contents: Introduction to GANs: Generator and Discriminator structure, Adversarial loss, Minimax optimization. Types of GANs: Vanilla GAN, Deep Convolutional GANs (DCGAN), Conditional GANs, Progressive Growing GAN. Challenges in GAN Training: Mode collapse, vanishing gradients.

Further Readings: Style Transfer and Creativity in Design: StyleGAN, CycleGAN

UNIT 4	Transformer-based Generative Models	08 HOURS
Applications/Case Study: OpenAI GPT-3, DALL-E and DALL-E 2, BERT. Contents: Introduction to Transformers: Self-attention mechanism, Positional encoding. Transformer-based Generative Models: GPT (GPT-2, GPT-3), BERT for generative tasks. Text-to-Image Synthesis, Vision Transformers (ViT) and generative applications in image generation. Further Readings: Creative Industries: Content Creation in Art, Music, and Design.		
UNIT 5	Diffusion Models	06 HOURS
Applications/Case Study: Medical Imaging Enhancement, Automotive Design and Simulation Contents: Introduction to Denoising Diffusion Probabilistic Models (DDPM). Producing high-quality, diverse outputs in image, video, and text generation Iterative and probabilistic approach: Forward diffusion (adding noise) and reverse process (denoising) Self Study: Multimodal Models		
UNIT 6	Ethical, Societal, and Legal Dimensions of Generative AI	06 HOURS
Applications/Case Study: Medical Imaging Enhancement, Automotive Design and Simulation Contents: Ethics: Fairness, transparency, accountability, bias mitigation, privacy, and human-AI collaboration in global and Indian contexts. Society: Misinformation, job displacement, cultural diversity, trust restoration, and applications in education and healthcare worldwide and in India. Policy Law: EU AI Act, General Data Protection Regulation, India's IT Act, Data Protection Bill, copyright, intellectual property, liability, and AI governance frameworks. Security Sustainability: Deepfakes, misuse prevention, energy efficiency, carbon footprint, Paris Agreement, and National AI Strategy (NITI Aayog).		

TEXT BOOKS

1. Jozefowicz, R., Zaremba, W., Sutskever, I. (2023). An Introduction to Generative Models with Applications. Cambridge University Press.
2. Choi, H., Kim, J. (2023). Generative AI: A Practical Guide to Applications and Ethics. Springer.
3. Liu, L., Xu, H. (2020). Generative deep learning: Teaching machines to paint, write, compose, and play. O'Reilly Media.
4. Goodfellow, I., Bengio, Y., Courville, A. (2016). Deep learning. MIT Press.
<https://www.deeplearningbook.org/>

REFERENCE BOOKS

1. Ramesh, A., Pavlov, M., Goh, G., et al. (2022). DALL·E: Creating images from text. OpenAI. Retrieved from <https://openai.com/research/dall-e>
2. Lee, J., Kim, J. (2023). Generative AI for Software Engineering: Techniques and Tools. CRC Press.

E-LEARNING RESOURCES

1. Swayam Course on Generative AI and Large Language Models, By Naveen Kumar Bhansali, Indian Institute of Management Bangalore (IIMB).
2. OpenAI. (n.d.). Research and blog on AI advancements. Retrieved from <https://www.openai.com/blog>.
3. Coursera. (n.d.). Generative adversarial networks (GANs) specialization by DeepLearning.AI. Coursera. Retrieved from <https://www.coursera.org>

 MIT Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)		COURSE SYLLABUS	
SCHOOL OF COMPUTER ENGINEERING		W.E.F	AY: 2024- 2025 (Rev. 2022)
FINAL YEAR BACHELOR OF TECHNOLOGY COMPUTER ENGINEERING		COURSE NAME	Generative Artificial Intelligence LAB
		COURSE CODE	2304422L
		COURSE CREDITS	1
RELEASE DATE : 01/07/2025		REVISION NO.	0.0

TEACHING SCHEME (HOURS/WEEK)		EXAMINATION SCHEME AND MARKS					
		THEORY			LABORATORY		TOTAL
LECTURE	PRACTICAL	IA	MSE	ESE	CA	PRACT/DEMO/PRES.	
NIL	2	NIL	NIL	NIL	20	30	50

PREREQUISITE KNOWLEDGE: Machine Learning, Deep Learning.

COURSE OBJECTIVES :

2304422L.CEO.1: Develop practical skills in building and deploying generative AI models.
2304422L.CEO.2: Analyze ethical considerations and impacts of generative AI in different fields.
2304422L.CEO.3: Explore advanced models such as Generative Adversarial Networks , Variational Autoencoders, Diffusion Models, and Transformer-based architectures.

COURSE OUTCOMES :

After successful completion of the course, students will be able to,

2304422L.CO.1: Analyze Generative AI models and diffusion models to generate data across various domains like images, text, and music [L4].
2304422L.CO.2: Evaluate the performance of Generative AI models for real world applications [L5].
2304422L.CO.3: Deploy generative AI for practical applications such as image synthesis, data augmentation, and creative content generation [L6].

COURSE ABSTRACT:

This advanced course in Generative AI explores the theoretical underpinnings and practical implementations of AI models capable of generating original content. Leveraging foundational knowledge from Artificial Intelligence and Machine Learning and Deep Learning, students will delve into the architecture, training, and application of generative models, including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Transformer-based models such as GPT.

Through hands-on projects and case studies, students will learn to design, implement, and evaluate generative models for tasks such as image synthesis, text generation, and style transfer. Emphasis is placed on understanding the ethical implications and potential biases of generative AI, as well as methods to mitigate risks in applications. By the end of the course, students will have practical expertise in deploying generative models and insights into emerging applications across industries.

PRACTICAL EXPERIMENTS

PRACT 1	Introduction to Generative AI	04 HOURS
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Practical Experiment:

1. Setting Up the Environment for Generative AI and Make use of generative models for creative content generation.
2. Implement simple probabilistic models like Gaussian Mixture Models.

PRACT 2	Autoencoders and Variational Autoencoders (VAE)	04 HOURS
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Practical Experiment

1. Build a simple AE model for Dimensionality Reduction and Denoising.
2. Generate realistic faces or interpolate between facial features for creative applications in entertainment and design using VAE.

PRACT 3	Generative Adversarial Networks (GANs)	04 HOURS
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Practical Experiment: Implement a DCGAN and/or cGAN on a chosen dataset to generate original image data, providing a comprehensive summary of the GAN architectures, training processes, and generated results. Conduct a comparative analysis of the models, evaluating their effectiveness in image quality and consistency. Conclude with a presentation that includes a case study analysis, discussing ethical considerations and challenges in real-world GAN applications.

PRACT 4	Transformer-based Generative Models	04 HOURS
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Practical Experiment:

1. Generate coherent and contextually relevant text using GPT models.
2. Create images based on text prompts using a text-to-image model to explore Transformer capabilities in multimodal tasks.

PRACT 5	Diffusion Models	04 HOURS
Practical Experiment: Implement a basic diffusion model for image generation by simulating the process of adding and removing noise. Use a pre-trained diffusion model to generate images from text prompts and compare results, highlighting model strengths, limitations, and ethical considerations.		
PRACT 6	Generative AI in Action: The Capstone Experience	10 HOURS
This Lab Experiment serves as the culmination of the Generative AI course, where students apply their acquired knowledge and skills to develop a capstone project that demonstrates innovative use of generative models. Students will work in teams to ideate, design, and implement a project that addresses a real-world challenge or explores a creative application of generative AI. Learning Outcomes: 1. Formulate a project proposal based on identified needs or creative ideas in the field of Gen AI. 2. Collaborate effectively in teams to execute a Gen AI project. 3. Implement, Analyze and Evaluate a Gen AI model using appropriate methodologies and tools. 4. Communicate project outcomes through a detailed report/article and a professional presentation.		

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3. Coursera. (n.d.). Generative adversarial networks (GANs) specialization by DeepLearning.AI. Coursera. Retrieved from <https://www.coursera.org>

